

Oral Supplementation of *Majun Baladar* Ameliorates Antioxidant Enzyme Activities in Cerebral Ischaemic Damage

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Abstract: *Majun Baladar* (MB), a traditional herbal formulation of the Unani system of medicine, was studied for its efficacy against cerebral ischaemia-induced oxidative damage in hippocampus and associated neurobehavioural deficits. Adult male Wistar rats were divided into four groups. The first group was sham, the second group was ischaemic (MCAO: middle cerebral artery occluded) and the third group was a MB pre-treated ischaemic group (MCAO + MB). The fourth group was given MB (1.05 g/kg) orally for 15 days as a drug control. The middle cerebral artery was occluded for 2 hr and reperused for 22 hr in the ischaemic as well as the drug pre-treated group. The activity of the various enzymatic antioxidants like glutathione peroxidase, glutathione reductase, glutathione S-transferase and non-enzymatic antioxidants, glutathione along with levels of lipid peroxidation were evaluated. Cerebral ischaemic rats showed elevated level of lipid peroxidation and decreased levels of various antioxidants significantly over sham values. As a result of MB pre-treatment, the level of lipid peroxidation was found to be significantly depleted as compared to the ischaemic group. Furthermore, depleted levels of glutathione and the activity of glutathione peroxidase, glutathione S-transferase and glutathione reductase were restored significantly in MB treated group. *Majun Baladar* exhibited a significant improvement in neurobehavioural activities in the drug pre-treated animals as compared to the ischaemic group as evidenced by the grip strength test, Rota-Rod and video path analysis. The results of the present study provide baseline information regarding the neuroprotective efficacy of MB and also open a window for a potent therapeutic use of this traditional herbal Unani medicine.

The brain is extremely sensitive to reduction in oxygen supply. Hippocampal neurones are very sensitive to ischaemic damage and are associated with memory impairments [1]. During the ischaemic cascade, the brain cells become vulnerable to various assaults, because parts of the cellular antioxidant defence systems are compromised that leads to depletion of cellular energy sources, release of excitatory amino acids, mitochondrial dysfunction, excessive generation of free radicals, all of which are contributing factors to the oxidative damage [2]. Limited endogenous antioxidant defences are likely to be ineffective due to the excess of oxygen species, inactivation of detoxification systems and degradation of antioxidants [3].

The ischaemic core is surrounded by *penumbra* that is electrically silent but still has significant blood flow and is the site of major therapeutic intervention. Therapeutic potential of new herbal compounds with antioxidant potential is a promising approach against oxidative damage. Modern research is driven by evidence-based medicine, whereas herbal medicine relies on centuries of experience with treatment using traditional formulations [4]. Traditional Unani medicine has long been used to treat various nervine disorders,

but there are very few scientific proofs on their therapeutic action. We have previously reported very interesting findings involving neuroprotective efficacy of various Unani medicines and herbs against cerebral ischaemia and Parkinson's disease. Yousuf et al. [5] reported the neuroprotective effect of a Unani formulation *Khamira Abresham Uood Mastagi-wala* against cerebral ischaemia-induced biochemical alterations in discrete brain parts of rat. Saleem et al. [6,7] evaluated the protective effect of *Nardostachys jatamansi* (herb) and herbal extract of saffron (*Crocus sativa*) against cerebral ischaemic damage in rats. *Nardostachys jatamansi* was also shown effective in the treatment of Parkinson's disease [8]. Some other herbs like *Delphinium denudatum*, *Ginkgo biloba*, *Withania somnifera*, Crocetin [9–12] and the Unani preparation *Khamira Abresham Hakim Arshad Wala* were reported to inhibit neuronal damage in Parkinson's disease [13].

On the basis of above mentioned findings, we were prompted to study one of the herbal Unani medicine formulations, *Majun Baladar* (MB). Avicenna (Ibn Sina in Arabic, AD 980–1037) [14] mentioned the pertinent qualities of MB in his prized book 'Canon of Medicine' (Al-Qanoon in Arabic). *Majun Baladar* has 19 constituents with their own pharmacological importance like saffron (*Crocus sativus*), honey, ginger, musk, almond, marking nut, ground nut, long pepper and sesame oil that diversifies their use in multiple therapeutic interventions (table 2). The neuroprotective aspect of MB has not yet been studied in a rat focal cerebral

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ischaemic model. Therefore, the purpose of this study was to determine the neuroprotective effects of MB against cerebral ischaemia-induced oxidative damage in hippocampus and the associated neurobehavioural deficits.

Materials and Methods

Drugs and chemicals. Glutathione (oxidized and reduced), nicotinamide adenine dinucleotide phosphate reduced form (NADPH), 1-chloro-2,4-dinitrobenzene (CDNB), 5,5'-dithiobis-2-nitrobenzoic acid (DTNB) and thiobarbituric acid (TBA) were purchased from Sigma-Aldrich (St. Louis, MO, USA). Other chemicals were of analytical reagent grade. *Majun Baladar* was procured from Hamdard (Waqf) Laboratories (Ghaziabad, Uttar Pradesh, India). The purity of MB was maintained by the manufacturers according to the standard scale (www.hm.hamdard.com).

Animals. Male Wistar rats, weighing 300–350 g, were used for the study as they can withstand stress more efficiently and give a clearer picture of behavioural outputs after stroke. Rats were obtained from the Central Animal House of Jamia Hamdard, New Delhi, India. They were housed in polypropylene cages in air-conditioned room and allowed free access to pellet diet and water *ad libitum*. The studies were carried out in accordance with the official regulations approved by the Animal Ethics Committee of Jamia Hamdard, New Delhi, India.

Experimental protocol. The animals were divided into four groups ($n = 6$). The first group served as sham and saline was given orally. The second group was the ischaemic group (MCAO: middle cerebral artery occluded) in which ischaemia was induced for 2 hr followed by reperfusion for 22 hr. The third group was pre-treated with MB (1.05 g/kg, orally for 15 days, single dose daily) followed by ischaemia/reperfusion (MCAO + MB group). The fourth group was pre-treated with MB alone for 15 days as a drug control. After the last dose of MB was given on day 15, ischaemia/reperfusion was induced on the same day. After reperfusion, the animals were assessed for neurobehavioural activities and then killed. The human dose of MB was converted to the animal dose (specifically for rat) and was found to be 1.05 g/kg that was used in the present investigation.

Induction of ischaemia. The right middle cerebral artery occlusion was produced using an intraluminal filament model as earlier described by Longa et al. [15]. In brief, the rats were anaesthetized with chloral hydrate (400 mg/kg, intraperitoneally). A 4–0 nylon monofilament pre-coated with poly-L-lysine was inserted into the external carotid artery and advanced into the internal carotid artery. Two hours after the induction of ischaemia, the filament was slowly withdrawn and animals were then returned to their cages for a period of 22 hr of reperfusion. In the sham-operated rats, the external carotid artery was surgically prepared for insertion of the filament but the filament was not inserted. The body temperature of animals was maintained with an infra-red lamp and was continuously monitored for respiration and unresponsiveness throughout the surgical procedure.

Behavioural study. The behavioural tests in each group were performed after producing occlusion and reperfusion. The experiment was performed between 9 a.m. and 4 p.m. at standard laboratory conditions. All tests were performed and analysed by a person blind to the experiment.

Physical performance tests.

Grip strength. Grip strength was performed according to the method of Moran et al. [16]. The apparatus with a string of 50 cm length, pulled out between two vertical supports and elevated according to the following scale: 0 – fall off; 1 – hangs onto string by two forepaws; 2 – as for 1 but attempts to climb on string; 3 – hangs onto string by two forepaws plus one or both hind limbs; 4 – hangs onto string

by forepaws plus tail wrapped around string and 5 – escape. The highest reading of three successive trials was taken for each animal.

Rota-Rod (motor coordination). Omni rotor (Omnitech Electronics Inc., Columbus, OH, USA) was used to evaluate the motor incoordination. It consisted of rotating rod (75 mm diameter), which was divided into four compartments, permitting test of four rats at a time. The apparatus automatically recorded time 0.1 sec. when the rats fall off the rotating shaft. The speed was set at 10 r.p.m. and cut off time was 180 sec.

Open field tests (locomotor activity). Observations were recorded for 10 min. according to the method of Yamamoto et al. [17] with slight modification at ambient temperature ranging between 26–32°C. Mean of session totals of vehicle and treatment groups were compared for locomotion (sec.), rest (sec.), distance travelled (cm), average speed (cm/sec.). Videopath analyser (Coulbourn Instrument, Allentown, PA, USA) consists of an open field chamber (50 × 50 × 35 cm), and a video camera was fixed over the chamber by an adjacent rod, an activity monitor, a programmer and a printer. Sessions total for all parameters were taken. Observations were recorded for 10 min.

Tissue preparation. After producing ischaemia and assessment of behavioural parameters, the animals were killed immediately and their brains were taken out to dissect hippocampus. Post-mitochondrial supernatant obtained from 10% homogenate of tissue in phosphate buffer (0.1 M, pH 7.4) was used for the estimation of various biochemical estimations.

Biochemical estimations.

Lipid peroxidation. The procedure of Utley et al. [18] was used for the estimation of the rate of lipid peroxidation with some modifications. One millilitre of homogenate (2.5% in chilled KCl) was pipetted in a 20 ml glass tube and incubated at $37 \pm 1^\circ\text{C}$ in a metabolic shaker for 60 min. Similarly, 1 ml of the same homogenate was pipetted in a centrifuge tube and incubated at 0°C . After 1 hr of incubation, 1.0 ml 5% chilled tetrachloroacetic acid was added followed by 1.0 ml of 0.67% TBA in each vial and proper mixing was done after each addition. The aliquot from each vial was transferred to a centrifuge tube and centrifuged at $2000 \times g$ for 15 min. Thereafter, the supernatant was transferred to another tube and placed in the boiling water bath. After 10 min., the test tubes were cooled and the absorbance of each aliquot was measured at 535 nm. The rate of lipid peroxidation was expressed as nanomoles of TBA reactive substance formed/hr/g of tissue using a molar extinction coefficient of $1.56 \times 10^5/\text{M}/\text{cm}$.

Glutathione (GSH). Reduced glutathione was determined by the method of Jollow et al. [19]. Post-mitochondrial supernatant was precipitated with sulfosalicylic acid (4.0%) in the ratio of 1:1. The sample were kept at 4°C for 1 hr and then centrifuged at $1200 \times g$ for 15 min. at 4°C . The assay mixture contained supernatant, 0.1 M phosphate buffer and DTNB (stock = 100 mM in 0.1 M phosphate buffer) in a total volume of 3 ml. The optical density of reaction product was read at λ 412 nm and results were expressed as nmol GSH/g tissue.

Estimation of total thiols. The assay was done according to the method of Sedlak and Lindsay [20]. The assay mixture contained 0.2 ml of homogenate (10% w/v), 0.1 M Tris-EDTA (pH 8.2) and 1.0 mM DTNB in total volume of 2.7 ml and mixed well. Thereafter, 8.3 ml methanol was added and centrifuged at $3500 \times g$ for 5 min. The yellow colour developed was read immediately at 412 nm in spectrophotometer (UV-1601, Shimadzu, Japan). The total thiol content was calculated by using molar extinction co-efficient of $13.6 \times 10^4/\text{M}/\text{cm}$.

Glutathione peroxidase (GPx). The activity of GPx was assayed according to the method described by Mohandas et al. [21]. The reaction mixture (2 ml) consisted of 1.44 ml sodium-phosphate buffer (0.1 M, pH 7.4), 0.1 ml EDTA (1 mM), 0.1 ml of 1 mM sodium azide,

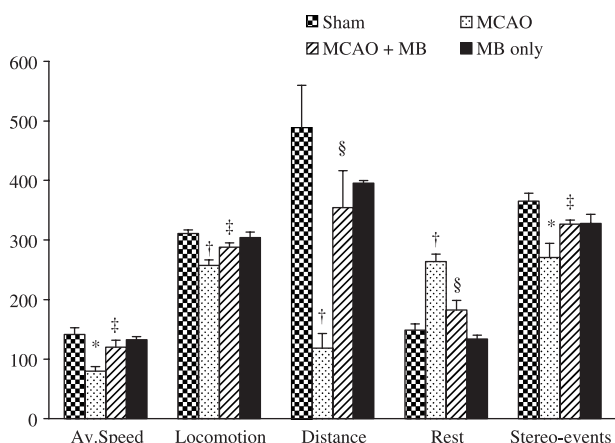


Fig. 1. Effect of *Majun Baladar* (MB) on open-field activities (average speed, locomotion, distance travelled, rest time and stereo events). Values are expressed as mean $\pm$ S.E. (n = 6). Significant changes were depicted as *P < 0.01, †P < 0.001 when compared to sham; ‡P < 0.05, §P < 0.01 when compared to middle cerebral artery-occluded (MCAO) group.

50 μ l glutathione reductase (1 IU/ml), 0.1 ml of 1 mM GSH, 0.1 ml NADPH (0.2 mM), 0.1 ml of 10% PMS and 10 μ l of 0.25 mM H_2O_2 . The enzyme activity was calculated as nmol NADPH oxidized/min./mg protein, using a molar extinction co-efficient of 6.22×10^3 M/cm.

Glutathione S-transferase (GST). Assay for the activity of GST was measured as described by Habig et al. [22]. For the assay of GST with CDNB as the substrate, the reaction mixture (2 ml) consisted of phosphate buffer (0.1 M, pH 7.4), 1 mM GSH, 10 mM CDNB and 10% post-mitochondrial supernatant. An increase in optical density at 340 nm was recorded every 30 sec. for 3 min. The results were calculated by using the molar extinction co-efficient of 9.6×10^3 M/cm for CDNB conjugate at 340 nm. The enzyme activity was expressed as nmol of CDNB conjugate formed/min./mg protein.

Glutathione reductase (GR). The activity was measured by the method of Carlberg and Mannervik [23]. Briefly, the reaction mixture (2 ml) consisted of 1.60 ml of phosphate buffer (0.1 mM, pH 7.4), 0.1 ml NADPH (0.1 mM), 0.1 ml EDTA (0.5 mM), 0.1 ml GSSG (1 mM) and 0.1 ml of post-mitochondrial supernatant (10%). The reaction mixture was incubated at 30°C for 5 min. before adding post-mitochondrial supernatant. The enzyme activity was determined by measuring the disappearance of NADPH at 340 nm using an extinction co-efficient 6.22×10^3 M/cm. Specific activity of the enzyme was expressed as NADPH oxidized/min./mg protein.

Catalase. Catalase was estimated by the method of Claiborne et al. [24]. The assay mixture consisted of 1.95 ml of 0.1 M phosphate buffer (pH 7.4), 50 μ l of post-mitochondrial supernatant (10%). The reaction was initiated by adding 1.0 ml of 0.019 M hydrogen peroxide. Decrease in optical density at 240 nm was recorded after every 30 sec. for 3 min. Enzyme activity was calculated in terms of nmol H_2O_2 consumed/min./mg protein.

Statistical analysis of data. Data were analysed for significance of difference using two-factor ANOVA, followed by the individual group analysis using the Student's t-test. Values are expressed as means $\pm$ S.E. (n = 6). The significance of results was ascertained at P < 0.05.

Results

Behavioural outputs.

Animals from MB group did not show any significant change in behaviour output as compared to the sham group.

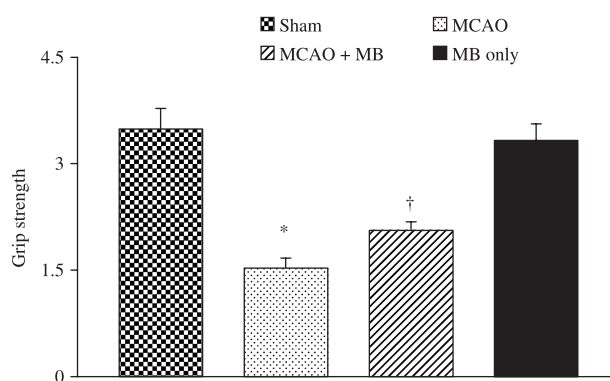


Fig. 2. Effect of *Majun Baladar* (MB) on grip strength. Values are expressed as mean $\pm$ S.E. (n = 6). Significance was determined as *P < 0.01 when compared to sham; †P < 0.05 when compared to middle cerebral artery-occluded (MCAO) values.

As a result of cerebral ischaemia, open-field studies showed a significant (P < 0.01–0.001) decrease in average speed, locomotion, distance travelled, stereo-events and a significant (P < 0.001) increase in rest time in the MCAO group when compared to the sham group (fig. 1). It was observed that animals from the MCAO + MB group showed a significant (P < 0.05–0.01) improvement in open-field studies as compared to the MCAO group (fig. 1). When a grip strength test was performed in different groups, a significantly (P < 0.01) weak grip strength was observed in the MCAO group as compared to the sham group. Whereas the MB pre-treatment was found to attenuate the weakened grip strength significantly (P < 0.05) in the MCAO + MB group when it was compared to the MCAO group (fig. 2). The results obtained from the Rota-Rod test showed a significant (P < 0.05) alteration in motor activity in the MCAO group as compared to the sham group. As a result of MB pre-treatment a significant (P < 0.05) improvement in balance and coordination was observed in the MCAO + MB group as compared to the MCAO group (fig. 3).

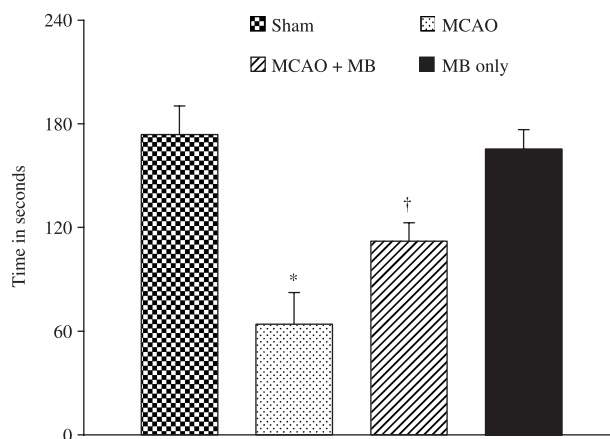


Fig. 3. Effect of *Majun Baladar* (MB) on muscular activity (Rota-Rod). Values are expressed as mean $\pm$ S.E. (n = 6). Significance was determined as *P < 0.01 when compared to sham; †P < 0.05 when compared to middle cerebral artery-occluded (MCAO) group.

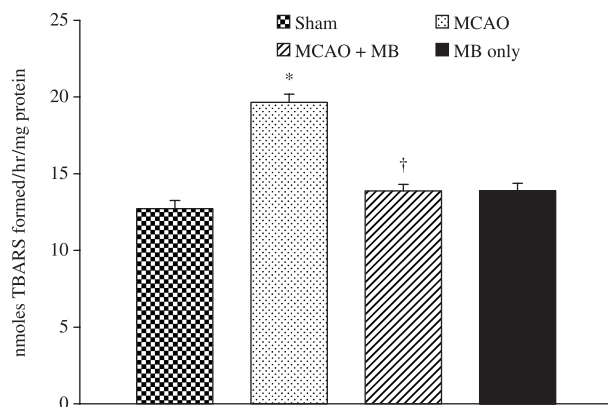


Fig. 4. Level of lipid peroxidation in hippocampus of various groups. Values are expressed as mean $\pm$ S.E. (n = 6). Significant difference * $P < 0.001$ when compared to sham; † $P < 0.001$ when compared to middle cerebral artery-occluded (MCAO) group. Unit of lipid peroxidation was expressed as nmols thiobarbituric acid reactive substance (TBARS) formed/hr/mg protein.

Lipid peroxidation.

A significant ($P < 0.001$) increase in the level of lipid peroxidation was found in hippocampus of the MCAO group as compared to the sham group (fig. 4). As a result of the MB pre-treatment, a marked decrease ($P < 0.01$) in the level of lipid peroxidation was observed in MCAO + MB group when compared to the MCAO group. Pre-treatment with MB alone did not show any significant change in the level of lipid peroxidation over sham values.

GSH and total thiol.

As a result of cerebral ischaemia, the cellular contents of GSH and total thiols was found to be decreased significantly ($P < 0.001$) in the hippocampus of the MCAO group as compared to sham values (figs 5 and 6). MB pre-treatment alone did not alter the hippocampal levels of either GSH or total thiols when compared to the sham group.

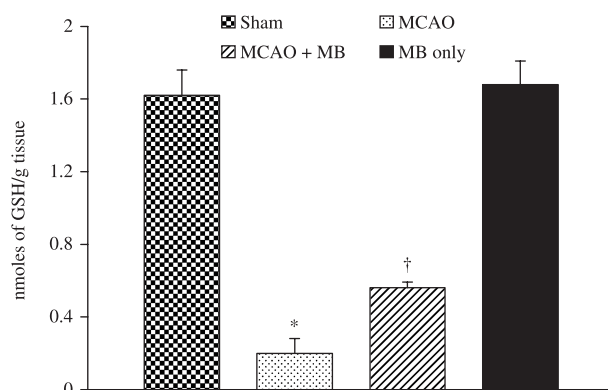


Fig. 5. Cellular glutathione (GSH) content in hippocampus of various groups. Values are expressed as mean $\pm$ S.E. (n = 6). Significant difference * $P < 0.001$ when compared to sham; † $P < 0.001$ when compared to middle cerebral artery-occluded (MCAO) group. Unit of GSH was expressed as expressed as nmols GSH/g tissue.

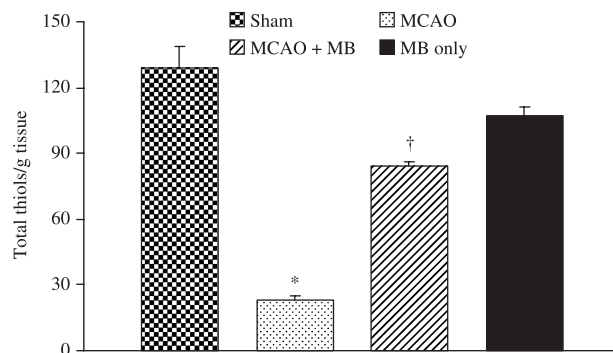


Fig. 6. Level of total thiols in hippocampus of various groups. Values are expressed as mean $\pm$ S.E. (n = 6). Significance was depicted as * $P < 0.001$ when compared to sham values; † $P < 0.001$ when compared to middle cerebral artery-occluded (MCAO) group. Unit of total thiols was expressed as total thiols/g tissue.

Pre-treatment with MB showed a significant increase ($P < 0.01$) in the level of GSH and total thiols in the MCAO + MB group as compared to the MCAO group.

Antioxidant enzymes.

Table 1 shows the effect of MB pre-treatment on the activities of various antioxidant enzymes of hippocampus from various groups. A significant ($P < 0.001$) decrease in the activities of GPx, GR, GST and catalase was observed in the MCAO group as compared to the sham group. No significant change was observed in the activity of various antioxidant enzymes in the MB alone group as compared to sham values. As a result of MB pre-treatment, a significant ($P < 0.01$) restoration was observed in the activity of various enzymes in the MCAO + MB group when compared to the MCAO group.

Discussion

In the present study, MB pre-treatment for 15 days protected against cerebral ischaemia-induced oxidative damage in hippocampus with improved neurobehavioural outcome. Although MB was also tested for its neuroprotective effect for 3 and 7 days, no satisfactory results were obtained (data not shown). Lipid peroxidation is one of the important markers of oxidative damage in the ischaemic cascade as polyunsaturated fatty acids constitute the major brain components. In the present study, the level of lipid peroxidation increased significantly with simultaneous decrease in GSH content in the ischaemic group. The results indicate that peroxidative stress takes its toll on the GSH content. Many other workers have reported increased level of lipid peroxidation and reduced cellular GSH content in brain tissue after cerebral ischaemia [5,6,25]. In addition, the activity of various antioxidant enzymes was found to be severely depleted in ischaemic animals, due to the exhaustion of cellular antioxidants during energy depletion as a result of cerebral ischaemia. There are several reports showing decrease in the antioxidant defence in the ischaemic brain

Table 1.

Activity of various antioxidant enzymes in hippocampus.

Enzymes	Sham	MCAO	MCAO + MB	MB only
GPx (nmole NADPH oxidized/min./mg protein)	18.75 ± 0.43	7.59 ± 0.38* (↓59.52%)	12.73 ± 0.68** (↑67.72%)	17.64 ± 0.82
GR (nmole NADPH oxidized/min./mg protein)	27.75 ± 0.34	19.48 ± 0.46* (↓29.80%)	23.18 ± 0.34** (↑18.99%)	26.55 ± 0.42
GST (nmole CDNB conjugate formed/min./mg protein)	15.63 ± 0.62	9.87 ± 0.51* (↓36.85%)	11.63 ± 0.21*** (↑17.83%)	14.67 ± 0.72
CAT (nmole H ₂ O ₂ consumed/min./mg protein)	6.38 ± 0.21	3.23 ± 0.09* (↓49.37%)	5.07 ± 0.57*** (↑56.96%)	6.13 ± 0.27

Values are expressed as mean ± S.E. Significant difference *P < 0.001 when compared to the sham group; **P < 0.001, ***P < 0.01 when compared to the MCAO group. Values in parentheses show the percentage increase or decrease with respect to their control values. CAT, catalase; CDNB, 1-chloro-2,4-dinitrobenzene; GPx, glutathione peroxidase; GR, glutathione reductase; GST, glutathione S-transferase; MB, *Majun Baladar*; MCAO, middle cerebral artery occluded; NADPH, nicotinamide adenine dinucleotide phosphate.

after cerebral ischaemia/reperfusion [26,27]. Yousuf et al. [5] reported a significant decrease in the activities of antioxidant enzymes and cellular GSH content in discrete parts of the ischaemic brain.

Interestingly, as a result of MB pre-treatment for 15 days, a different picture was observed in the drug pre-treated ischaemic group. It was observed that the level of lipid peroxidation was depleted to a notable level with the increase in cellular GSH and total thiols content in the MB pre-treated ischaemic group. The activities of various antioxidant enzymes were also restored. This turn in the responses may be attributed to the ingredients of MB with their own pharmacological importance. To name a few: saffron (*Crocus sativus*), honey, ginger, mastich, almond, marking nut, ground nut, long pepper and sesame that are all rich sources of antioxidants and capable of scavenging free radicals [28–36]. Ikeda et al. [37] reported that supplementation of sesame seed (sesamin and sesaminol) elevated the α -tocopherol levels and subsequently decreased the lipid peroxidation level and red blood aemolysis in the plasma and liver of rats. Deng et al. [38] suggested that carotenoids present in saffron act as a membrane-associated high-efficiency free radical scavenger and increase blood flow due to vasodilation, which is useful in treating ischaemic neuropathy. Saffron is also reported to increase the intracellular levels of GSH and GSH-dependent antioxidant enzymes in rat brain after cerebral ischaemia [7]. Furthermore, in the present study, MB pre-treatment reduced lipid peroxidation levels and increased cellular GSH and total thiol content in ischaemic animals. This might be due to the lipid peroxidation lowering effect of sesame oil and free radical scavenging activity of the carotenoids present in saffron.

In the present study, we observed that the MB pre-treated ischaemic group showed a significant increase in the depleted activity of various antioxidant enzymes. How MB led to the restoration of antioxidant enzymes is not clear; however, the individual pharmacological aspects of the various ingredients with potent antioxidant properties might have acted to reinstate the depleted enzyme activities by scavenging free radicals. There are many reports suggesting

the antioxidant potential of natural compounds that are present in the formulation of MB (table 2).

In addition, the antioxidant properties of the active components of saffron have been described. One of the important active components is crocetin that has been reported to play a protective role in a hemi-Parkinsonism model of rats by restoration of various antioxidant enzyme activities, decrease in dopamine and lipid peroxidation content that was confirmed by histological studies [12]. Ginger is another important ingredient of MB that has been reported to possess superoxide and hydroxyl radical scavenging activity [44]. In the ischaemic cascade, superoxide and hydroxyl radicals are the basic perpetrators of oxidative damage [53]. In the present study, the protective effect of MB against ischaemia/reperfusion-induced depletion in antioxidant enzymes could be attributed to the presence of ginger. Hippocrates (460–357 BC) identified the wound healing and the various antioxidants properties of honey. The presence of pinocembrin, pinobanksin, chrysin, galagin, ascorbic acid (vitamin C), catalase and selenium adds to shield the free radical-mediated neuronal damage [54]. This antioxidant property of honey could also have added to the protective role of MB in the present study.

In response to cerebral ischaemia, various behavioural impairments are well-documented in Wistar rats e.g. memory, locomotion and muscle strength dysfunctions [5,6,27]. The efficacy of various neuroprotectants to the improved behavioural outcome may be considered as an instant basic response of any therapeutic intervention. In the present study, the MB pre-treated ischaemic group showed improved motor activity, grip strength and also distance travelled, etc. when compared to the control group. It can be argued that the reduction in blood flow and blood oxygen might alter the functions of the motor regions. Our results here clearly demonstrate that not only does MB ameliorate the altered biochemical states but also improves neurological and behavioural outcome. This effect of MB can be attributed to the vasodilatory activity of carotenoids present in saffron that might have increased the blood flow and oxygen content to the affected regions.

In conclusion, oxidative stress is among the myriad perpetrators of cerebral ischaemia. *Majun Baladar* afforded

Table 2.

Constituents of *Majun Baladar* (MB).

SN	Name (Arabic/Urdu)	Name (English)	Origin	Content (w/kg)	Clinical importance
1	<i>Kunjad Mukshir (Til)</i>	Sesame (<i>Sesamum indicum</i>)	Plant	4 g	Strengthens body, throat infections, relaxant, brain vitalizer [39], neuroprotective [36].
2	<i>Magaz Tukhm Baladar</i>	Marking nut (<i>Semecarpus anacardium</i>)	Plant	3 g	Skin disorders, increases blood circulation, general body vitalizer, used in nerve disorders (e.g. paralysis, epilepsy, hallucinations, important brain vitalizer [39], antitumour, cardioprotective, neurological disorders [33]).
3	<i>Magaz Badam</i>	Almond (<i>Prunus</i>)	Plant	3 g	Golden standard as memory enhancer, brain vitalizer, body vitalizer, cough, cold, anti-inflammatory, heart vitalizer [39], lowers cholesterol [32].
4	<i>Magaz Chilgoza</i>	Ground-nut (<i>Arachis hypogaea</i> L.)	Plant	3 g	Energizer, expectorant, used in epilepsy, paralysis [39], antioxidant [34].
5	<i>Aqir Qarha</i>	Pyrethrium root	Plant	3 g	Anaesthetic effect, gingivitis, nerve tonic, used in paralysis, facial palsy, epilepsy, tetanus [39].
6	<i>Kholnajan</i>	Galangal (<i>Alpinia officinarum</i>)	Plant	3 g	Strengthens body, used in sore throat [39], antioxidative properties [40], anti-inflammatory properties.
7	<i>Basbasa (Javitri)</i>	<i>Myristica fragans</i>	Plant	3 g	Relaxant, digestive, laxative, in stomach ailments, strengthens body, headache [39], antidepressant [41], anticarcinogenic activity [42].
8	<i>Jazboa (Jaiphal)</i>	Nutmeg	Plant	2 g	Relaxant, strengthens heart, memory enhancer, strengthens body, stimulates circulation, digestive system and anti-inflammation [39].
9	<i>Zanjabeel (Adrak)</i>	Ginger (<i>Zingiber</i>)	Plant	2 g	Strengthens body, digestive, strengthens brain, memory enhancer, laxative, expectorant, antispasmodic [39], antioxidant, mitigation of radiation-induced emesis [43], Effective in treating Parkinsonism [44], antioxidant and antityrosinase activity [30].
10	<i>Salab Misri</i>	Salep	Plant	2 g	Strengthens body, nutritive [39].
11	<i>Falfal Daraz</i>	Long pepper (<i>Piper longum</i>)	Plant	2 g	Analgesic, used in cough, asthma, whooping cough, stomach disorders [39], treatment of bronchial diseases, memory enhancer, aphrodisiac, neuroprotective effects of its gum extract may be due to increasing antioxidant defence and suppression free radicals production in brain [35].
12	<i>Mastagi</i>	Mastich	Plant	2 g	Stimulant, catarrh, astringent [39], decreased iron-induced lipid peroxidation [31].
13	<i>Tukhm Gazar</i>	Carrot (<i>Daucus carota</i> L.)	Plant	1 g	Analgesic, nerve/heart disorders [39], hepatoprotective, cognitive dysfunctions, lowers cholesterol, anticholinesterase activity [45].
14	<i>Tukhm Konch</i>	Kapikachu/Cowhage (<i>Mucuna Prurines</i>)	Plant	1 g	General body enhancer, contains L-dopa, anti-Parkinson's drug [46], antihelminthic, diuretic, emollient, stimulant, aphrodisiacum, purgative, febrifuge, tonic, neuropathy, elephantiasis, ulcers, delirium [47].
15	<i>Zafran</i>	Saffron (<i>Crocus sativa</i> L.)	Plant	1 g	Anti-inflammatory, antiseptic, cardiac tonic, cough, cold [39], recovery from ischaemia/reperfusion injury [28], improves learning and memory [48], prevents tumour formation, atherosclerosis, hepatic damage, antioxidant [49], improves cerebral oxygenation in haemorrhaged rats [50].
16	<i>Samandar Sookh</i>	Sage (<i>Salvia plebeian</i>)	Plant	5 g	Urinary tract infections, Dioscorides (40–90 CE) reported that it stopped bleeding; Pliny the Elder (23–79 CE) reported its enhanced memory functions [51].
17	<i>Mushk</i>	Musk	Animal	6 mg	Protective, strengthens important body organs, paralysis, exhilarant [39].
18	<i>Kand safad</i>	Sugar	Plant	208.33 g	Additive
19	<i>Honey-ponay (Shahed)</i>	Honey	Animal	751.664 g	Energizer, antioxidant, digestive, increases blood circulation, expectorant, laxative, sucralfate-like substance, affords protection against ischaemia-reperfusion-induced injuries [29], bacteriostatic, bactericidal and antimicrobial [52].

neuroprotection against cerebral ischaemia/reperfusion-mediated oxidative damage in hippocampus and associated neurobehavioural deficits due to the synergistic antioxidant effect of the 19 constituents. The present study provides part of the scaffolding to construct a logical platform for the appearance of a new scientific discipline in this unexplored area.

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